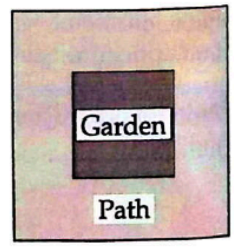
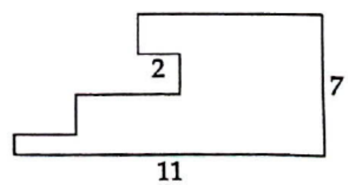
4.2.6★ The perimeter of a square garden is 64 meters. The path surrounding the garden has uniform width and has an area of 228 square meters. How many meters of fencing are needed to surround the outer edge of the path? *(Source: MATHCOUNTS)* **Hints: 494**



4.31 All sides of the building shown at right meet at right angles. If three of the sides measure 2 meters, 7 meters, and 11 meters as shown, then what is the perimeter of the building in meters? *(Source: Mandelbrot)*



4.33 The diagram at right above is formed by placing 4 squares together along a single line as shown. Each square has a side length that is 1/2 the side length of the next larger square. The outer perimeter of the figure is 115. What is the area of the whole figure? *(Source: ARML)* **Hints:** 329

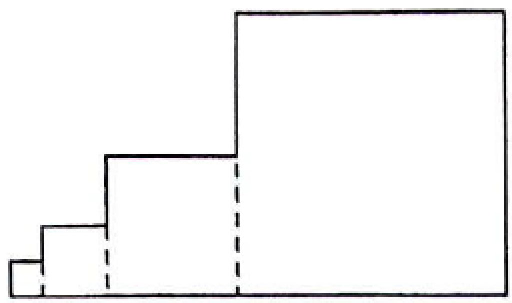


Figure 4.12: Diagram for Problem 4.33

4.34 Sue and Barry are trying to find the area of $\triangle ABC$. Sue mistakenly uses AB and the height from A (instead of the height from C), and Barry mistakenly uses BC and the height from C (instead of the height from A).

(a) Sue finds an area of 12 and Barry finds an area of 27. What is the area of $\triangle ABC$? **Hints:** 369, 38

4.35 Point P is on $\overline{WY}$ in the diagram at left below. Show that $[WPX] = [WPZ]$. **Hints:** 426, 108

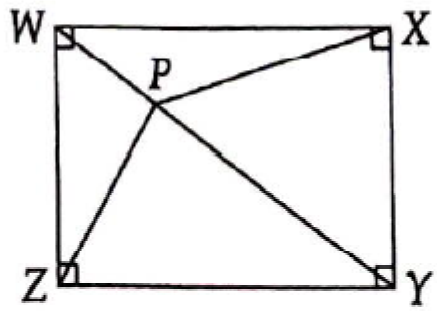


Figure 4.13: Diagram for Problem 4.35

4.38★ In the diagram at left below, given $[PQRS] = 5[PQA]$ and $[PQRS] = 4[PBS]$, find $[ABP]/[PQRS]$. **Hints:** 583, 470, 361

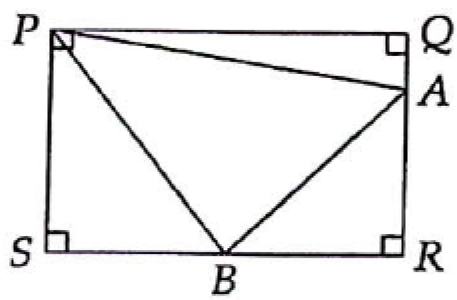


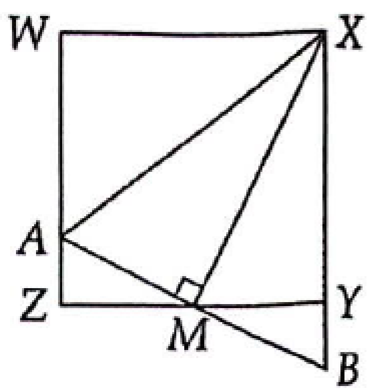
Figure 4.15: Diagram for Problem 4.38

4.40★ For Joe's birthday, Will has bought a 7×7 inch square birthday cake with a flat top. It is a chocolate cake with banana frosting on the top and on the sides. It turns out that seven people will be present when the cake is cut, and each person will become quite envious if another person receives more cake or frosting. Find a way to divide the cake among seven people, so that each receives an equal amount of cake and frosting. **Hints:** 149, 67, 124

5.1.2 $\triangle ABC \sim \triangle ADB$, $AC = 4$, and $AD = 9$. What is AB? *(Source: MATHCOUNTS)* **Hints:** 113

5.2.3 In the diagram, $WXYZ$ is a square. M is the midpoint of $\overline{YZ}$, and $\overline{AB} \perp \overline{MX}$.

(d) Prove that $\triangle AZM \sim \triangle MYX$, and use this fact to prove $AZ = XY/4$.

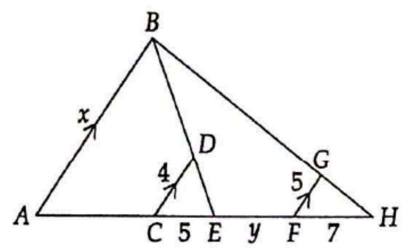


5.2.4 In triangle ABC, $AB = AC$, $BC = 1$, and $\angle BAC = 36^\circ$. Let D be the point on side $\overline{AC}$ such that $\angle ABD = \angle CBD$.

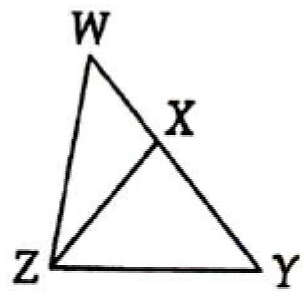
(a) Prove that triangles ABC and BCD are similar.

(b)★ Find AB. **Hints:** 150

5.2.5★ Find x in terms of y given the diagram below. **Hints:** 258, 522



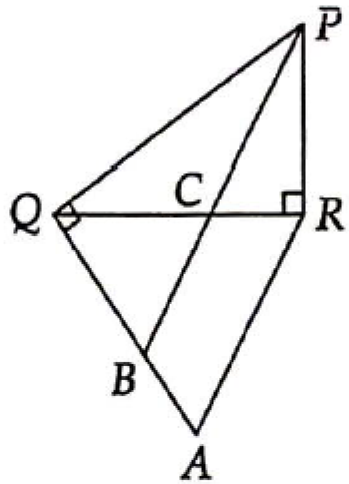
5.3.3 Show that if $WZ^2 = (WX)(WY)$ in the diagram at left below, then $\angle WZX = \angle WYZ$. **Hints:** 147



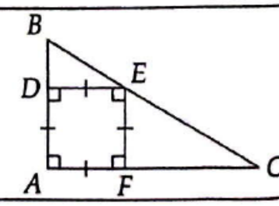
5.3.4 In the diagram at right above, $\angle PRQ = \angle PQA = 90°$, $QR = QA$, and $\angle QPC = \angle RPC$.

(a) Prove $\angle QCB = \angle QBC$. **Hints:** 202

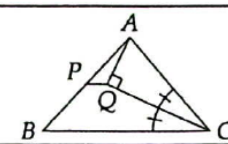
(b)★ Prove $\overline{RA} \parallel \overline{PB}$. **Hints:** 388



Problem 5.13: As shown in the diagram, $\angle A = 90°$ and $ADEF$ is a square. Given that $AB = 6$ and $AC = 10$, find AD.

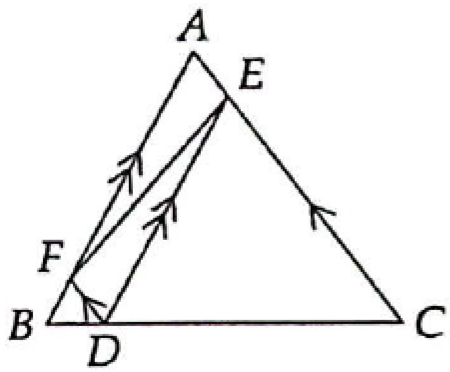


Problem 5.16: In the diagram, $\angle ACQ = \angle QCB$, $\overline{AQ} \perp \overline{CQ}$, and P is the midpoint of $\overline{AB}$. Prove that $\overline{PQ} \parallel \overline{BC}$. **Hints:** 35, 417



5.5.2 In the figure, the area of $\triangle EDC$ is 25 times the area of $\triangle BFD$.

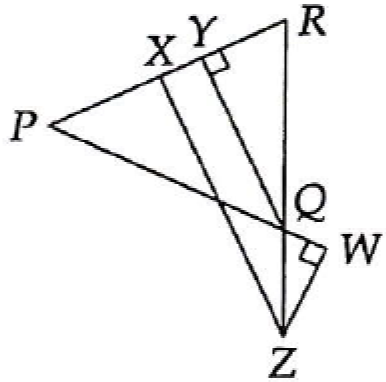
(c)$\star$ Find $[AFE]/[ABC]$. **Hints:** 321



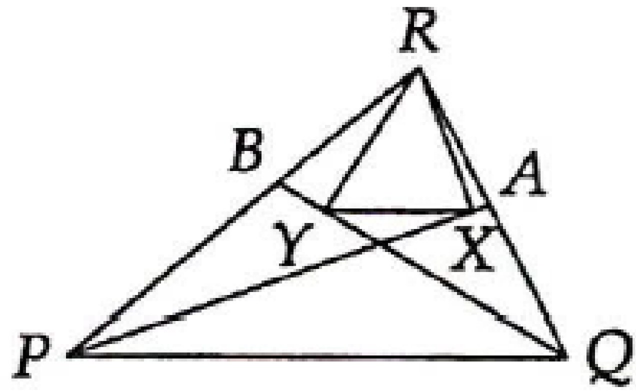
5.5.4 In the diagram at left below, $PQ = PR$, $\overline{ZX} \parallel \overline{QY}$, $\overline{QY} \perp \overline{PR}$, and $\overline{PQ}$ is extended to W such that $\overline{WZ} \perp \overline{PW}$.

(a) Show that $\triangle QWZ \sim \triangle RXZ$. **Hints:** 360

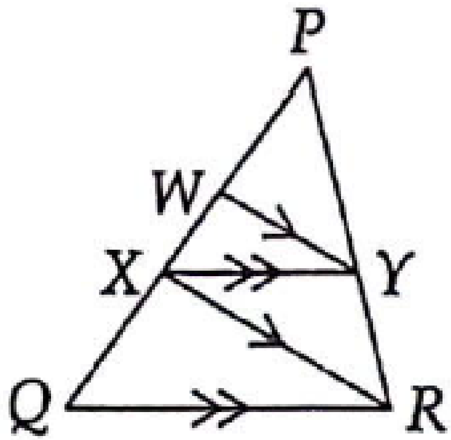
(b)$\star$ Show that $YQ = ZX - ZW$. **Hints:** 172, 550



5.5.5★ $\overline{PA}$ and $\overline{BQ}$ bisect angles $\angle RPQ$ and $\angle RQP$, respectively. Given that $\overline{RX} \perp \overline{PA}$ and $\overline{RY} \perp \overline{BQ}$, prove that $\overline{XY} \parallel \overline{PQ}$. **Hints:** 584, 152, 254



5.33 In the diagram at right above, $PW = 6$ and $WX = 4$. Find QX.



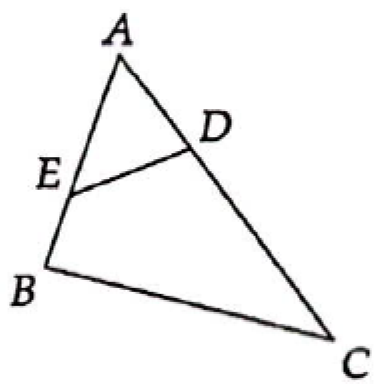
5.36 Let ABC be a triangle, and let D and E be points on sides $\overline{AB}$ and $\overline{AC}$, respectively, such that $\overline{DE} \parallel \overline{BC}$. Prove that

$$\frac{AD}{AE} = \frac{DB}{CE}.$$

5.37 Let ABC be a triangle, and let D and E be points on $\overline{AB}$ and $\overline{AC}$, respectively, such that $AD/AE = BD/EC$. Prove that $\overline{DE} \parallel \overline{BC}$. *Make sure you see why this differs from the previous problem!* **Hints:** 363, 179

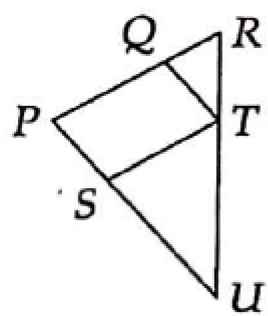
5.38 If the sum of one of the base angles and the vertex angle is the same for two different isosceles triangles, must the triangles be similar? **Hints:** 196

5.40 In triangle ABC at right above, D and E are points on sides $\overline{AC}$ and $\overline{AB}$, respectively, such that $(AD)(AC) = (AE)(AB)$. Prove that $\angle CDE + \angle CBE = 180°$ and $\angle ADB + \angle BEC = 180°$. **Hints:** 269, 70

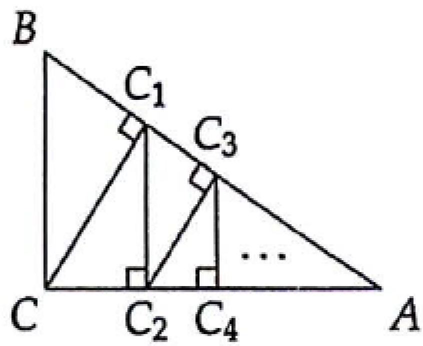


5.41 D, E, and F are on sides $\overline{BC}$, $\overline{AC}$, and $\overline{AB}$, respectively, of $\triangle ABC$ such that $\overline{DE} \parallel \overline{AB}$, $\overline{DF} \parallel \overline{AC}$, and $\overline{BC} \parallel \overline{EF}$. Prove that D, E, and F are the midpoints of the sides of $\triangle ABC$. **Hints: 24**

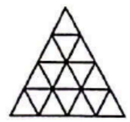
5.42 In the diagram at left below, $\overline{PS} \parallel \overline{QT}$ and $\overline{PQ} \parallel \overline{ST}$. Prove that $SU/SP = QP/QR$.



5.46★ In triangle ABC at right above, $AC = 4$, $BC = 3$, $AB = 5$, and $\angle ACB = 90^\circ$. The infinite sequence of points $C_1, C_2, C_3, C_4, \ldots$ is generated as follows: C_1 is the foot of the altitude from C to side $\overline{AB}$, C_2 is the foot of the altitude from C_1 to side $\overline{AC}$, C_3 is the foot of the altitude from C_2 to side $\overline{AB}$, and so on. Calculate the sum $CC_1 + C_1C_2 + C_2C_3 + C_3C_4 + \cdots$. **Hints:** 138, 307

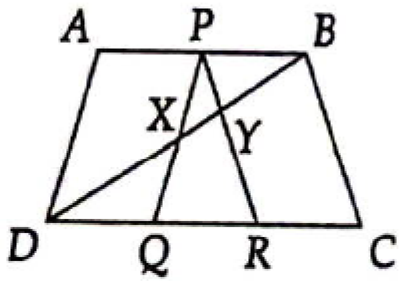


5.47★ In the figure shown, we have taken an equilateral triangle and divided each side into four segments of equal length. We have then connected these points to form smaller equilateral triangles.

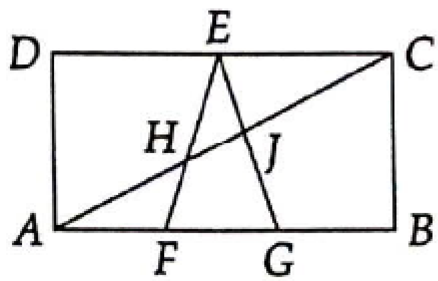


Consider instead dividing each side into n segments of equal length, where n is some positive integer, then connecting these points as before to form smaller equilateral triangles. Use this dissection of the original equilateral triangle to prove that the sum of the first n positive odd integers is n^2. **Hints:** 367, 475

5.48★ Figure $ABCD$ at left below has sides $AB = 6$, $CD = 8$, $BC = DA = 2$, and $\overline{AB} \parallel \overline{CD}$. Segments are drawn from the midpoint P of $\overline{AB}$ to points Q and R on side $\overline{CD}$ so that $\overline{PQ}$ and $\overline{PR}$ are parallel to $\overline{AD}$ and $\overline{BC}$, as shown. Diagonal $\overline{DB}$ intersects $\overline{PQ}$ at X and $\overline{PR}$ at Y. Evaluate PX/YR. *(Source: Mandelbrot)*



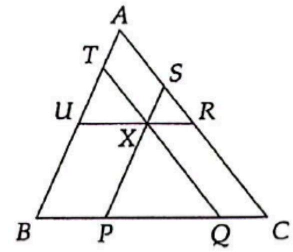
5.49★ In rectangle $ABCD$ at right above, points F and G lie on $\overline{AB}$ so that $AF = FG = GB$ and E is the midpoint of $\overline{DC}$. Also, $\overline{AC}$ intersects $\overline{EF}$ at H and $\overline{EG}$ at J. The area of rectangle $ABCD$ is 70. Find the area of $\triangle AHF$. *(Source: AMC 12)* **Hints:** 257, 319, 393



5.50★ Given an acute triangle $\triangle ABC$, construct with straightedge and compass square $DEFG$ such that D and E are on $\overline{BC}$, G is on $\overline{AB}$, and F is on $\overline{AC}$. **Hints:** 201, 132

5.51★ Points P, Q, R, S, T, and U are on the sides of triangle ABC, as shown, such that line segments $\overline{UR}$, $\overline{QT}$, and $\overline{SP}$ all pass through point X, and are parallel to $\overline{BC}$, $\overline{CA}$, and $\overline{AB}$, respectively. Prove that

$$\frac{PQ}{BC} + \frac{RS}{CA} + \frac{TU}{AB} = 1.$$

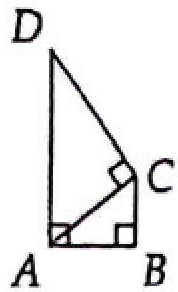


Hints: 353, 223, 261

6.1.10★ A triangle has sides measuring 13 cm, 13 cm, and 10 cm. A second triangle is drawn with sides measuring 13 cm, 13 cm and x cm, where x is a whole number other than 10. If the two triangles have equal areas, what is the value of x? *(Source: MATHCOUNTS)* **Hints:** 161, 326

6.2.4 Find the area of $\triangle PQR$ given that $PQ = QR = 8$ and $\angle PQR = 120^\circ$. **Hints:** 312

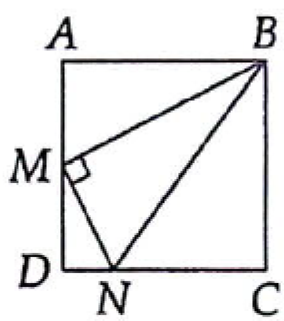
6.4.4 $\overline{AD}$ and $\overline{BC}$ are both perpendicular to $\overline{AB}$ in the diagram at left below, and $\overline{CD} \perp \overline{AC}$. If $AB = 4$ and $BC = 3$, find CD. *(Source: HMMT)*



6.4.5 $ABCD$ is a square in the diagram at right above. M is the midpoint of $\overline{AD}$ and $\overline{BM} \perp \overline{MN}$.

(a) Prove that $\triangle MDN \sim \triangle BAM$.

(b)★ Prove that $\angle ABM = \angle MBN$. **Hints:** 189

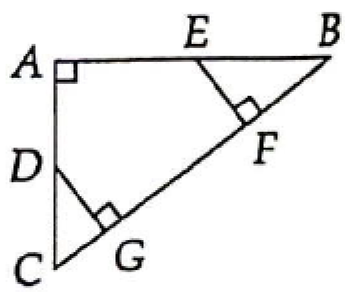


6.33 Find the perimeter of an isosceles triangle with base length 10 and area 60.

6.35 In the figure shown, E is the midpoint of $\overline{AB}$, D is the midpoint of $\overline{AC}$, $AB = 16$, and $AC = 12$. *(Source: MATHCOUNTS)*

(a) Show that $\triangle EFB \sim \triangle CAB$.

(b) Find DG and EF.



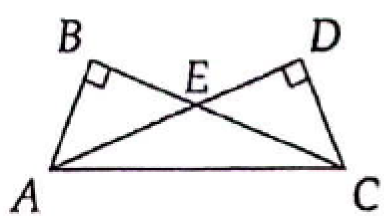
6.41 The length of one leg of a right triangle is 22, and the other two sides also have integer lengths. Find the perimeter of the triangle.

6.42 Point A is on side $\overline{XZ}$ of $\triangle XYZ$ such that $XA = XY$. Given that $\angle YXZ = 90°$, $\angle YZX = 30°$, and $ZA = 6 - \sqrt{12}$, find the area of $\triangle XYZ$.

6.44 A right triangle has area 210 and hypotenuse 29. Find the perimeter of the triangle.

6.48 In the diagram at right $\angle B = \angle D = 90^\circ$, $AB = DC = 24$, and $BC = AD = 32$. *(Source: ARML)*

(c) Find the area of $\triangle AEC$. **Hints:** 296



6.49 Triangle ABC has $AB = 12$, $BC = 16$, and $AC = 20$. If D is on $\overline{AC}$ such that $AD = 12$, find the area of $\triangle ADB$. *(Source: ARML)* **Hints:** 116, 424

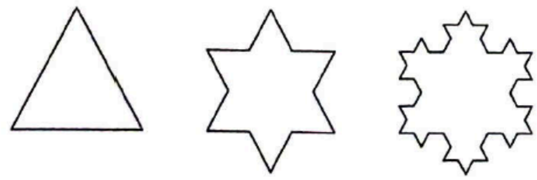
6.50 Riders on a Ferris wheel travel in a circle in a vertical plane. A particular wheel has radius 20 feet and revolves at the constant rate of one revolution per minute. How many seconds does it take a rider to travel from the bottom of the wheel to a point 10 vertical feet above the bottom? *(Source: AMC 10)*

6.54★ If the lengths of the three sides of a right triangle are whole numbers, how many such distinct noncongruent right triangles exist having one leg with a length of 24 units?

6.55 In $\triangle RST$, $RS = 13$, $ST = 14$, and $RT = 15$.

(b)★ Let M be the midpoint of $\overline{ST}$. Find RM. **Hints:** 563

6.56★ We begin with an equilateral triangle with side length 1. We divide each side into three segments of equal length, and add an equilateral triangle to each side using the middle third as a base. We then repeat this, to get a third figure.



If we continue this process forever, what is the area of the resulting figure? **Hints:** 13, 398

6.57★ Let $\triangle XOY$ be a right-angled triangle with $\angle XOY = 90^\circ$. Let M and N be the midpoints of legs $\overline{OX}$ and $\overline{OY}$, respectively. Given that $XN = 19$ and $YM = 22$, find XY. *(Source: AMC 10)* **Hints:** 290, 505

6.58★ In triangle ABC, $AB = AC$, $BC = \sqrt{3} - 1$, and $\angle BAC = 30°$. Find the length of AB.

6.59★ In $\triangle ABC$, $AB = 6$ and $BC = 8$. Find AC in each of the following cases:

(a) $\angle B = 30°$. **Hints:** 57

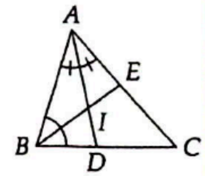
(b) $\angle B = 45°$. **Hints:** 129

(c) $\angle B = 135°$. **Hints:** 215

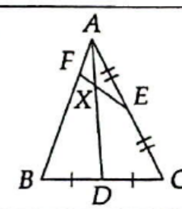
7.4.4★ Medians $\overline{XA}$ and $\overline{YB}$ of $\triangle XYZ$ have the same length. Prove that $XZ = YZ$. **Hints:** 167, 375

Problem 7.27: Given $AB = BC = 10$ and $AC = 12$, find the circumradius and the inradius of $\triangle ABC$.
Hints: 39, 322, 549

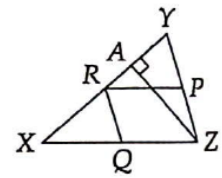
Problem 7.28: In the diagram, $\overline{AD}$ bisects $\angle BAC$ and I is the incenter of $\triangle ABC$. Furthermore, $AB = 7$, $BC = 8$, and $AC = 11$. Find AI/ID.



Problem 7.30: Point E is the midpoint of $\overline{AC}$, and $\overline{AD}$ is a median of $\triangle ABC$. F is on $\overline{AB}$ such that $AF = AB/4$. $\overline{EF}$ and $\overline{AD}$ meet at X. Find AX/AD. **Hints:** 95



7.6.1 P, Q, and R are the midpoints of sides $\overline{YZ}$, $\overline{XZ}$, $\overline{XY}$, respectively, and $\overline{ZA}$ is the altitude to side $\overline{XY}$.

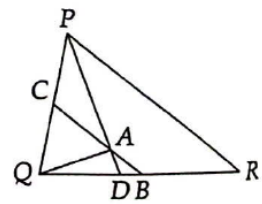


(a) Show that $AP = RQ$. **Hints:** 298

(b)★ Show that $\angle PRQ = \angle PAQ$. **Hints:** 47

7.6.2 In right triangle ABC, let F be the midpoint of hypotenuse $\overline{AB}$, and let D be the foot of the altitude from C to $\overline{AB}$. Let E be on $\overline{AB}$ such that $\overline{CE}$ is the angle bisector of $\angle ACB$. Prove that $\angle DCE = \angle ECF$.

7.6.3 In $\triangle PQR$, $PQ = 8$, $PR = 10$, and $QR = 9$. $\overline{PD}$ bisects $\angle QPR$, $\overline{QA} \perp \overline{PD}$, and $\overline{BC}$ passes through A such that $\overline{BC} \parallel \overline{PR}$.



(a) Prove that $\overline{AC}$ is a median of $\triangle PAQ$.

(b)★ Find DB. **Hints:** 532, 226, 125

7.37 Altitudes $\overline{AD}$ and $\overline{CF}$ of acute triangle $\triangle ABC$ meet at H. Prove that $\angle CHD = \angle ABC$.

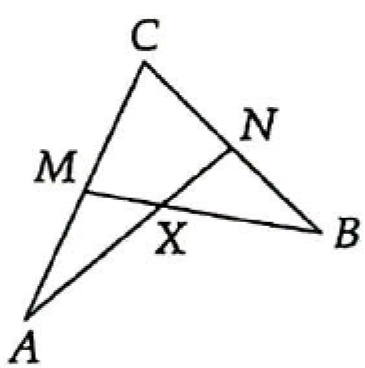
7.41 $\triangle TUV$ is isosceles with $TU = UV = 30$ and $TV = 36$. $\overline{UY}$ bisects $\angle TUV$ and $\overline{TM}$ is a median of $\triangle TUV$. $\overline{UY}$ and $\overline{TM}$ meet at X.

(e) Find XT and XM.

7.44 Point O is the circumcenter of $\triangle PQR$, $\angle QPR = 45°$, and $\angle QPO = 23°$.

(b)★ Find $\angle OQR$.

7.45 M and N are the midpoints of $\overline{CA}$ and $\overline{CB}$, respectively, as shown at right. $\overline{BM}$ and $\overline{AN}$ meet at X. Given that $XM = 3.5$ and $XA = 7.2$, find XN and XB.



7.46 Medians $\overline{AX}$ and $\overline{BY}$ of triangle $\triangle ABC$ are perpendicular at point O. $AX = 12$ and $BC = 10$.

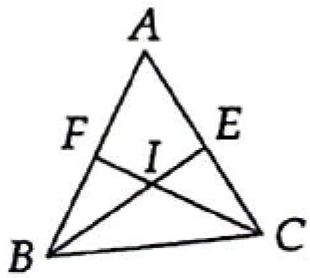
(a) Find AO and BY.

(b)★ Find the length of median $\overline{CZ}$.

7.47 Cevians $\overline{AD}$ and $\overline{BE}$ meet at point X inside $\triangle ABC$. $\overline{CX}$ extended meets $\overline{AB}$ at F. If $[AEX] = [CEX] = [CDX] = [BDX] = [BXF] = [AXF]$, must X be the centroid of $\triangle ABC$?

7.48 $\overline{BE}$ and $\overline{CF}$ are angle bisectors that meet at I as shown at right. $CE = 4$, $AE = 6$, and $AB = 8$.

(b) Find BC.



7.51 Let H and O denote the orthocenter and circumcenter of acute triangle ABC, respectively. $\overleftrightarrow{AH}$ meets $\overleftrightarrow{BC}$ at P and $\overline{AO}$ meets $\overline{BC}$ at Q. Prove that $\angle BAP = \angle CAQ$. **Hints:** 499, 288

7.52 In $\triangle JKL$, we have $JK = JL = 25$ and $KL = 30$. Find the following:

(c) the circumradius of $\triangle JKL$. **Hints:** 305

7.53 Let I be the incenter of triangle ABC, and let $\overrightarrow{AI}$ meet $\overline{BC}$ at A'.

(b)★ Prove that $A'I/IA = BC/(AB + AC)$. **Hints:** 372

7.54 In triangle ABC, $\angle A = 100^\circ$, $\angle B = 50^\circ$, and $\angle C = 30^\circ$. H is on $\overline{BC}$ and M on $\overline{AC}$ such that $\overline{AH}$ is an altitude and $\overline{BM}$ is a median. Find $\angle MHC$. *(Source: AHSME)* **Hints:** 464

7.56 In $\triangle ABC$, $\angle C$ is a right angle. Point M is the midpoint of $\overline{AB}$, point N is the midpoint of $\overline{AC}$, and point O is the midpoint of $\overline{AM}$. The perimeter of $\triangle ABC$ is 112 and $ON = 12.5$. What is the area of $MNCB$? *(Source: MATHCOUNTS)*

7.60★ A ladder is initially resting vertically against a wall that is perpendicular to the ground. It begins to slip and fall to the ground, with the bottom of the ladder moving directly from the wall, and the top of the ladder always touching the wall. What path does the midpoint of the ladder trace? (Don't forget to prove that the midpoint of the ladder really does hit every point on the path!) **Hints:** 271, 347

7.63★ Point N is on hypotenuse $\overline{BC}$ of $\triangle ABC$ such that $\angle CAN = 45°$. Given $AC = 8$ and $AB = 6$, find AN. **Hints:** 448, 560, 585

7.64★ The circle at right has radius 1 and is circumscribed about equilateral triangle ABC. If X is the midpoint of $\overline{AC}$ and Y is on arc $\widehat{AC}$ such that $\angle YXA$ is right, then find length XY. *(Source: Mandelbrot)*